TASK1

Query | Query History

```sql
-- 1. Белгілі авиакомпанияның барлық рейстері
SELECT f.*
FROM flights f
JOIN airline a ON a.airline_id = f.airline_id
WHERE a.airline_name = 'Air Astana';
```

Data Output | Messages | Notifications

Showing rows: 1 to 32 | Page No: 1 of 1

	flight_id [PK] integer	flight_no character varying (50)	scheduled_departure date	scheduled_arrival date	departure_airport_id integer	arrival_airport_id integer	departing_gate character varying (50)	arriving_gate character varying (50)
1	13	BR-PE	2024-01-16	2023-06-02	13	20	1891	5429
2	33	MZ-G	2023-09-21	2023-11-29	4	19	[null]	9380
3	36	AU-NT	2023-03-29	2023-05-06	12	14	[null]	13
4	73	FR-K	2023-12-26	2023-04-30	18	9	144	23
5	78	US-VT	2023-08-28	2023-08-02	20	11	335	515
6	143	PH-BUK	2023-08-20	2023-10-24	12	9	919	1330
7	170	SD-01	2023-12-02	2023-12-17	3	20	897	1147
8	175	PA-8	2023-03-19	2023-08-21	19	19	150	38
9	185	NA-KU	2023-06-26	2023-12-04	6	5	2000	996
10	221	US-CT	2023-11-07	2023-05-03	20	18	30	100
11	222	RU-CU	2023-04-16	2023-09-15	9	12	558	33
12	274	BE-VAN	2023-05-03	2023-06-01	13	6	53	57
13	318	US-AK	2024-03-01	2023-11-27	3	9	[null]	31
14	332	PG-NSB	2023-11-26	2023-12-05	8	20	230	13
15	346	US-MS	2023-03-28	2023-10-25	15	18	298	16
16	357	JP-41	2023-10-05	2023-12-01	12	6	6	714
17	446	KH-9	2023-06-12	2024-01-26	12	13	[null]	1295

TASK2

Query | Query History

```sql
-- 2. Барлық рейстер және ұшу аэропорттарының атаулары
SELECT f.flight_id, f.flight_no, ap.airport_name AS departure_airport
FROM flights f
JOIN airport ap ON ap.airport_id = f.departure_airport_id;
```

Data Output | Messages | Notifications

Showing rows: 1 to 10

	flight_id integer	flight_no character varying (50)	departure_airport character varying (50)
1	1	US-CT	Elorza Airport
2	2	US-NM	Figari Sud-Corse Airport
3	3	FI-OL	Darchula Airport
4	4	RU-KR	Lime Acres Finsch Mine Airport
5	5	RO-DJ	Hana Airport
6	6	CA-SK	Darchula Airport
7	7	AU-TAS	Ocean Falls Seaplane Base
8	8	US-AZ	Figari Sud-Corse Airport
9	9	IN-OR	Hana Airport
10	10	AU-NT	Longana Airport
11	11	TH-57	Delta County Airport
12	12	CA-NL	Elorza Airport
13	13	BR-PE	Figari Sud-Corse Airport
14	14	US-IN	Darchula Airport
15	15	ML-2	Akunaq Heliport
16	16	NA-CA	Zephyrhills Municipal Airport
17	17	ES-AN	Figari Sud-Corse Airport
18	18	TH-32	Fort Worth Alliance Airport

TASK3

Query | Query History

```
12  -- 3. Келесі айда рейсі жоқ авиакомпаниялар
13  SELECT a.airline_id, a.airline_name
14  FROM airline a
15  LEFT JOIN flights f
16         ON f.airline_id = a.airline_id
17        AND f.scheduled_departure >= date_trunc('month', CURRENT_DATE) + INTERVAL '1 month'
18        AND f.scheduled_departure <  date_trunc('month', CURRENT_DATE) + INTERVAL '2 month'
19  WHERE f.flight_id IS NULL;
```

Data Output | Messages | Notifications

Showing rows: 1 to 50 | Page No: 1

	airline_id [PK] integer	airline_name character varying (50)
1	20	SSL
2	25	CLY
3	26	YHB
4	27	KKL
5	11	IVA
6	39	GYA
7	17	YLP
8	33	BLD
9	31	YEK
10	34	SMM
11	12	KOQ
12	10	SOZ
13	18	KIQ
14	2	PDN
15	47	DNU

TASK4

Query | Query History

```
21  -- 4. Белгілі рейстегі жолаушылар тізімі
22  SELECT p.passenger_id, p.first_name, p.last_name
23  FROM passengers p
24  JOIN booking b         ON b.passenger_id = p.passenger_id
25  JOIN booking_flight bf ON bf.booking_id  = b.booking_id
26  JOIN flights f         ON f.flight_id    = bf.flight_id
27  WHERE f.flight_no = 'KC101';
28
```

Data Output | Messages | Notifications

	passenger_id [PK] integer	first_name character varying (50)	last_name character varying (50)

TASK5

Query | Query History

```
29  -- 5. Әр рейс бойынша билет бағасының орташа, жалпы, максимум, минимумы
30  SELECT f.flight_id, f.flight_no,
31         AVG(b.price) AS avg_price,
32         SUM(b.price) AS total_price,
33         MAX(b.price) AS max_price,
34         MIN(b.price) AS min_price
35  FROM flights f
36  JOIN booking_flight bf ON bf.flight_id  = f.flight_id
37  JOIN booking b         ON b.booking_id  = bf.booking_id
38  GROUP BY f.flight_id, f.flight_no;
```

Data Output | Messages | Notifications

Showing rows: 1 to 614

	flight_id [PK] integer	flight_no character varying (50)	avg_price numeric	total_price numeric	max_price numeric	min_price numeric
1	273	AU-WA	8929.2000000000000000	8929.20	8929.20	8929.20
2	951	PG-MPL	3034.5400000000000000	6069.08	5888.12	180.96
3	70	MY-12	683.2700000000000000	683.27	683.27	683.27
4	350	CN-41	5936.7680000000000000	29683.84	9792.28	2520.04
5	539	FR-F	5874.0900000000000000	17622.27	7927.75	3116.63
6	874	CA-MB	4901.8866666666666667	14705.66	7822.69	1839.80
7	176	AZ-YE	6895.9500000000000000	13791.90	9849.54	3942.36
8	292	VN-U-A	4500.8800000000000000	4500.88	4500.88	4500.88
9	663	UG-212	4316.5200000000000000	4316.52	4316.52	4316.52
10	770	IN-MP	1486.5900000000000000	2973.18	2104.71	868.47
11	417	GB-ENG	2410.0800000000000000	2410.08	2410.08	2410.08
12	556	GR-A1	3733.9833333333333333	11201.95	9224.38	416.30
13	764	PG-CPK	3347.0600000000000000	6694.12	3762.86	2931.26

TASK6

Query | Query History

```
40  -- 6. Белгілі елге ұшатын рейстер (flights + airport + airline)
41  SELECT f.flight_no, al.airline_name, ap.airport_name, ap.country
42  FROM flights f
43  JOIN airport ap ON ap.airport_id = f.arrival_airport_id
44  JOIN airline al ON al.airline_id = f.airline_id
45  WHERE ap.country = 'Kazakhstan';
46
```

Data Output | Messages | Notifications

	flight_no character varying (50)	airline_name character varying (50)	airport_name character varying (50)	country character varying (50)

TASK7

Query | Query History

```sql
-- 7. Кәмелетке толмаған жолаушылар және баратын жері
SELECT DISTINCT p.first_name, p.last_name, p.date_of_birth,
       ap.airport_name AS destination, ap.city, ap.country
FROM passengers p
JOIN booking b         ON b.passenger_id = p.passenger_id
JOIN booking_flight bf ON bf.booking_id  = b.booking_id
JOIN flights f         ON f.flight_id    = bf.flight_id
JOIN airport ap        ON ap.airport_id  = f.arrival_airport_id
WHERE p.date_of_birth > CURRENT_DATE - INTERVAL '18 years';
```

Data Output | Messages | Notifications

Showing rows: 1 to 17 | Page No: 1

	first_name character varying (50)	last_name character varying (50)	date_of_birth date	destination character varying (50)	city character varying (50)	country character varying (50)
1	Bernie	Michal	2009-11-03	Zephyrhills Municipal Airport	Cankova	Slovenia
2	Cleve	Edgeler	2009-04-20	Alert Bay Airport	Dubrava	Croatia
3	Cleve	Edgeler	2009-04-20	Fort Worth Alliance Airport	Rogóźno	Poland
4	Cleve	Edgeler	2009-04-20	Garbaharey Airport	Mlawe	Indonesia
5	Cleve	Edgeler	2009-04-20	Industrial Airpark	Guihulñgan	Philippines
6	Cleve	Edgeler	2009-04-20	Lime Acres Finsch Mine Airp...	Tielu	China
7	Ronnica	Laybourn	2009-03-06	Delta County Airport	Polykárpi	Greece
8	Ronnica	Laybourn	2009-03-06	Elorza Airport	Shuiting	China
9	Ronnica	Laybourn	2009-03-06	Garbaharey Airport	Mlawe	Indonesia
10	Ronnica	Laybourn	2009-03-06	Hana Airport	Wilmington	United States
11	Ronnica	Laybourn	2009-03-06	Industrial Airpark	Guihulñgan	Philippines
12	Vivyan	Mallabone	2009-11-01	Alert Bay Airport	Dubrava	Croatia
13	Vivyan	Mallabone	2009-11-01	Darchula Airport	Hilotongan	Philippines
14	Vivyan	Mallabone	2009-11-01	Elorza Airport	Shuiting	China
15	Vivyan	Mallabone	2009-11-01	Hana Airport	Wilmington	United States

TASK8

Query | Query History

```sql
-- 8. Толық аты, паспорт нөмірі, келудің ағымдағы (нақты) уақыты
SELECT p.first_name || ' ' || p.last_name AS full_name,
       p.passport_number,
       COALESCE(f.actual_arrival, f.scheduled_arrival) AS arrival_time
FROM passengers p
JOIN booking b         ON b.passenger_id = p.passenger_id
JOIN booking_flight bf ON bf.booking_id  = b.booking_id
JOIN flights f         ON f.flight_id    = bf.flight_id;
```

Data Output | Messages | Notifications

Showing rows: 1 to 1

	full_name text	passport_number character varying (50)	arrival_time date
1	Muhammad Fass	109932770-9	2023-07-18
2	Trevar Broun	788025864-7	2024-02-11
3	Auria Breffit	570537341-4	2023-07-11
4	Archie Toffel	677556708-1	2023-06-17
5	Sanders Biddles	514546405-3	2023-09-05
6	Sanders Biddles	514546405-3	2024-03-01
7	Remington Piggot	470074456-1	2023-05-31
8	Glynis Marle	209933120-0	2024-02-02
9	Cesaro McGennis	364900190-X	2023-04-03
10	Con Borrel	837859465-3	2024-02-15
11	Lorianna Robbie	435354132-3	2024-01-19

TASK9

Query | Query History

```
66  -- 9. Авиакомпания елі = ұшу елі болатын рейстер, аэропорт елі бойынша топтау
67  SELECT ap.country AS airport_country,
68         COUNT(*) AS flights_count,
69         STRING_AGG(f.flight_no, ', ' ORDER BY f.flight_no) AS flights
70  FROM flights f
71  JOIN airline al ON al.airline_id = f.airline_id
72  JOIN airport ap ON ap.airport_id = f.departure_airport_id
73  WHERE al.airline_country = ap.country
74  GROUP BY ap.country;
75
```

Data Output | Messages | Notifications

Showing rows: 1 to 8

	airport_country character varying (50)	flights_count bigint	flights text
1	Brazil	5	CL-AR, IT-45, PG-WPD, SB-WE, US-WA
2	China	50	AU-NSW, AU-NT, AU-WA, AU-WA, BR-AM, BR-BA, BR-MG, BR-PE, BS-NP, CA-BC, CA-ON,
3	Greece	1	AU-NT
4	Indonesia	6	CL-AN, CN-23, NZ-CAN, UG-212, US-AK, US-AK
5	Philippines	3	BR-PR, US-MI, US-MN
6	Poland	2	AU-QLD, VN-U-A
7	Russia	5	EE-39, IL-D, JP-12, MY-14, US-FL
8	Slovenia	2	CA-QC, PG-SAN

TASK10

Query | Query History

```
76  -- 10. Рейс нөмірі, авиакомпания, ұшу және қону аэропорттары
77  SELECT f.flight_no, al.airline_name,
78         dep.airport_name AS departure_airport,
79         arr.airport_name AS arrival_airport
80  FROM flights f
81  JOIN airline al   ON al.airline_id  = f.airline_id
82  JOIN airport dep  ON dep.airport_id = f.departure_airport_id
83  JOIN airport arr  ON arr.airport_id = f.arrival_airport_id;
84
```

Data Output | Messages | Notifications

Showing rows: 1 to 1000

	flight_no character varying (50)	airline_name character varying (50)	departure_airport character varying (50)	arrival_airport character varying (50)
1	US-CT	GYA	Elorza Airport	Ocean Falls Seaplane Base
2	US-NM	SMM	Figari Sud-Corse Airport	Zephyrhills Municipal Airport
3	FI-OL	SMM	Darchula Airport	Elorza Airport
4	RU-KR	BLD	Lime Acres Finsch Mine Airport	Longana Airport
5	RO-DJ	PIR	Hana Airport	Alert Bay Airport
6	CA-SK	SMM	Darchula Airport	Akunaq Heliport
7	AU-TAS	SOZ	Ocean Falls Seaplane Base	Darchula Airport
8	US-AZ	KOQ	Figari Sud-Corse Airport	Hana Airport
9	IN-OR	IFH	Hana Airport	Pitalito Airport
10	AU-NT	NHT	Longana Airport	Zephyrhills Municipal Airport
11	TH-57	YBQ	Delta County Airport	Delta County Airport
12	CA-NL	KKL	Elorza Airport	Figari Sud-Corse Airport
13	BR-PE	Air Astana	Figari Sud-Corse Airport	Melilla Airport

TASK11

Query | Query History

```
85  -- 11. Белгілі авиакомпания рейстеріне билет алған жолаушылар
86  SELECT p.first_name || ' ' || p.last_name AS full_name, f.flight_no
87  FROM passengers p
88  JOIN booking b          ON b.passenger_id = p.passenger_id
89  JOIN booking_flight bf ON bf.booking_id  = b.booking_id
90  JOIN flights f          ON f.flight_id    = bf.flight_id
91  JOIN airline al         ON al.airline_id  = f.airline_id
92  WHERE al.airline_name = 'Air Astana';
93
```

Data Output | Messages | Notifications

Showing row

	full_name text	flight_no character varying (50)
1	Jodi Ramm	ID-RI
2	Sheela Roux	AU-NT
3	Cesaro McGennis	FR-F
4	Rock Manuaud	US-TX
5	Hayward Keeler	US-MS
6	Brad Apperley	US-MS
7	Marillin Jewess	CA-BC
8	Karlens Mitro	US-IN
9	Samaria Carletto	KH-9
10	Sanders Biddles	AE-AZ
11	Paquito Dunrige	US-VT
12	Marvellen MacGilmar	IN-RJ

TASK12

Query | Query History

```
94  -- 12. Рейс нөмірі, баратын елі, жолаушы саны (брон жоқ рейстерді қоса)
95  SELECT f.flight_no, ap.country AS destination_country,
96         COUNT(bf.booking_id) AS passengers_count
97  FROM flights f
98  JOIN airport ap              ON ap.airport_id = f.arrival_airport_id
99  LEFT JOIN booking_flight bf ON bf.flight_id  = f.flight_id
100 GROUP BY f.flight_id, f.flight_no, ap.country;
101
```

Data Output | Messages | Notifications

Showing rows: 1 to 1

	flight_no character varying (50)	destination_country character varying (50)	passengers_count bigint
1	RU-ORL	Indonesia	1
2	RU-KR	China	0
3	US-WA	Poland	0
4	NA-OT	Greece	0
5	MY-14	Indonesia	2
6	CA-AB	Afghanistan	1
7	US-OK	China	0
8	PG-SAN	China	0
9	US-OH	Philippines	0
10	ZA-NP	Russia	2
11	AO-CCU	Croatia	1
12	CA-MB	Philippines	2

TASK13

Query | Query History

```
101
102  -- 13. Ешқандай билет алмаған жолаушылар
103  SELECT p.passenger_id, p.first_name, p.last_name
104  FROM passengers p
105  LEFT JOIN booking b ON b.passenger_id = p.passenger_id
106  WHERE b.booking_id IS NULL;
```

Data Output | Messages | Notifications

	passenger_id [PK] integer	first_name character varying (50)	last_name character varying (50)
1	82	Boote	Laurisch
2	31	Miquela	Tattersfield
3	146	Rudiger	Leith-Harvey
4	193	Sigfrid	Finding
5	123	Conn	Philipps
6	96	Weylin	Dubery
7	108	Fairfax	Brahmer
8	24	Osborn	Golsby
9	110	Der	Yanson
10	114	Andrew	Longmore
11	138	Ade	Goolden
12	190	Orrin	Billingham
13	16	Artur	Leap
14	148	Leonard	Wauchope
15	111	Luigi	Stove
16	144	Orville	Cable

TASK14

Query | Query History

```
109  -- 14. Каждая авиакомпания и общее число её рейсов (включая компании без рейсов)
110  SELECT a.airline_id,
111         a.airline_name,
112         COUNT(f.flight_id) AS total_flights
113  FROM airline a
114  LEFT JOIN flights f ON f.airline_id = a.airline_id
115  GROUP BY a.airline_id, a.airline_name
116  ORDER BY total_flights DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 50

	airline_id [PK] integer	airline_name character varying (50)	total_flights bigint
1	1	Air Astana	32
2	43	KMA	31
3	29	NHT	29
4	19	CSC	28
5	47	DNU	27
6	8	QIG	25
7	23	SPI	25
8	26	YHB	25
9	16	DUC	25
10	31	YEK	24
11	44	XLU	24
12	40	CII	24
13	6	NGL	23

TASK15

Query | Query History

```
118  -- 15. Рейсы, вылетающие из аэропортов указанной страны:
119  --      название авиакомпании, номер рейса, город аэропорта вылета
120  SELECT al.airline_name,
121         f.flight_no,
122         ap.city AS departure_city
123  FROM flights f
124  JOIN airline al ON al.airline_id  = f.airline_id
125  JOIN airport ap ON ap.airport_id  = f.departure_airport_id
126  WHERE ap.country = 'China';
```

Data Output | Messages | Notifications

Showing rows:

	airline_name character varying (50)	flight_no character varying (50)	departure_city character varying (50)
1	GYA	US-CT	Shuiting
2	BLD	RU-KR	Tielu
3	KKL	CA-NL	Shuiting
4	CSC	BR-TO	Tielu
5	HOU	CH-TI	Shuiting
6	KMA	UZ-SU	Luodian
7	YBQ	NO-18	Tielu
8	KGF	MN-1	Shuiting
9	DNU	US-AK	Luodian
10	CLY	US-CA	Sangzhou
11	Air Astana	AU-NT	Shuiting
12	KIQ	ID-MU	Linfen
13	SJS	PG-MPL	Shuiting
14	JKR	OM-ZA	Shuiting